

---

# Bayesian Latent Class Regression and Variable Selection with Applications to Sleep Patterns Data: Supplementary Material

---

## Contents

|                                                                        |           |
|------------------------------------------------------------------------|-----------|
| <b>S1 Simulation Studies</b>                                           | <b>1</b>  |
| S1.1 Simulation 1: Parameter Estimates and Selection Results . . . . . | 1         |
| S1.2 Simulation 2: Parameter Estimates and Selection Results . . . . . | 4         |
| <b>S2 CSHQ Sleep Data Analysis</b>                                     | <b>8</b>  |
| S2.1 Study Sample Characteristics . . . . .                            | 8         |
| S2.2 Model Parameter Estimates . . . . .                               | 8         |
| S2.3 Variable Selection Results . . . . .                              | 11        |
| S2.4 Supplementary Visualisations . . . . .                            | 12        |
| <b>S3 Technical Derivations</b>                                        | <b>14</b> |
| S3.1 Derivation of Pólya-Gamma Sampling Method for LCR . . . . .       | 14        |

## Simulation Studies

### Simulation 1: Parameter Estimates and Selection Results

Table S1 contains the true and estimated values of the item probability parameters for the full LCR model for simulation 1, computed by posterior mean from Gibbs sampling. Table S2 contains the estimates for the item probability parameters from the item selection model computed post-hoc. Table S3 contains posterior inclusion probabilities for each item variable in simulation 1. Figure S1 contains a ridgeline plot illustrating the posterior distributions of the regression coefficients for the full LCR model without variable selection applied to simulation 1.

| Var. | Cat. | True 1 | Est. 1 (sd) | True 2 | Est. 2 (sd) |
|------|------|--------|-------------|--------|-------------|
| 1    | 1    | 0.15   | 0.08 (0.03) | 0.70   | 0.74 (0.03) |
|      | 2    | 0.25   | 0.30 (0.03) | 0.20   | 0.16 (0.02) |
|      | 3    | 0.60   | 0.61 (0.04) | 0.10   | 0.10 (0.02) |
| 2    | 1    | 0.20   | 0.23 (0.03) | 0.55   | 0.55 (0.03) |
|      | 2    | 0.35   | 0.32 (0.03) | 0.30   | 0.29 (0.03) |
|      | 3    | 0.45   | 0.45 (0.04) | 0.15   | 0.15 (0.02) |
| 3    | 1    | 0.10   | 0.14 (0.03) | 0.80   | 0.82 (0.03) |
|      | 2    | 0.15   | 0.15 (0.03) | 0.15   | 0.13 (0.02) |
|      | 3    | 0.75   | 0.71 (0.04) | 0.05   | 0.05 (0.02) |
| 4    | 1    | 0.25   | 0.26 (0.03) | 0.45   | 0.44 (0.03) |
|      | 2    | 0.40   | 0.35 (0.04) | 0.35   | 0.37 (0.03) |
|      | 3    | 0.35   | 0.40 (0.04) | 0.20   | 0.19 (0.02) |
| 5    | 1    | 0.40   | 0.39 (0.04) | 0.40   | 0.44 (0.03) |
|      | 2    | 0.50   | 0.53 (0.04) | 0.50   | 0.45 (0.03) |
|      | 3    | 0.10   | 0.08 (0.02) | 0.10   | 0.11 (0.02) |
| 6    | 1    | 0.70   | 0.75 (0.03) | 0.70   | 0.72 (0.03) |
|      | 2    | 0.10   | 0.07 (0.02) | 0.10   | 0.09 (0.02) |
|      | 3    | 0.20   | 0.18 (0.03) | 0.20   | 0.19 (0.02) |
| 7    | 1    | 0.10   | 0.08 (0.02) | 0.10   | 0.09 (0.02) |
|      | 2    | 0.50   | 0.50 (0.04) | 0.50   | 0.52 (0.03) |
|      | 3    | 0.40   | 0.42 (0.04) | 0.40   | 0.38 (0.03) |
| 8    | 1    | 0.33   | 0.28 (0.03) | 0.33   | 0.32 (0.03) |
|      | 2    | 0.33   | 0.34 (0.03) | 0.33   | 0.34 (0.03) |
|      | 3    | 0.33   | 0.38 (0.04) | 0.33   | 0.34 (0.03) |

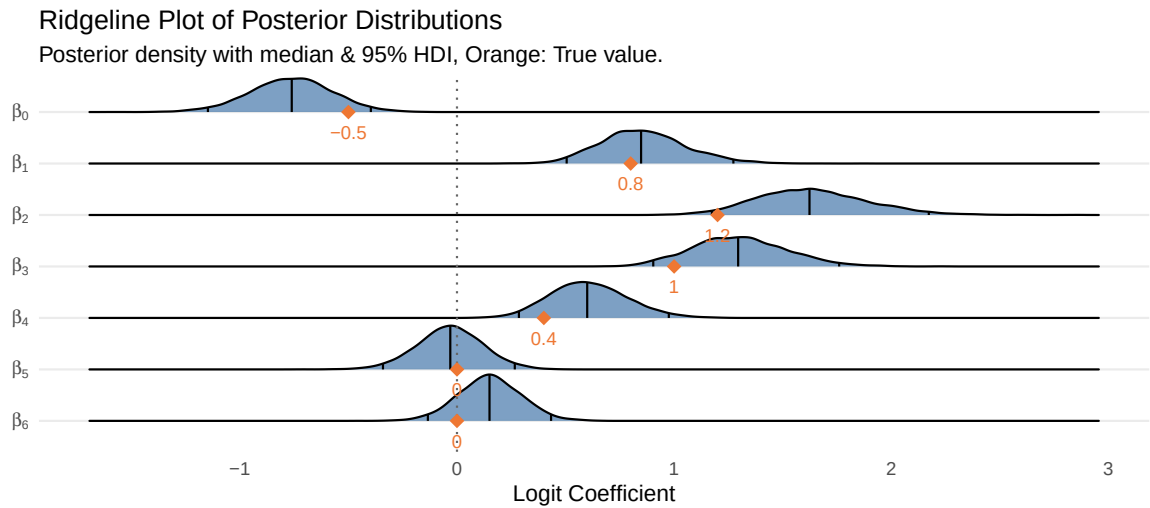
**Table S1.** True and estimated values of the item probability parameters for the full LCR model for simulation 1.

| Var. | Cat. | True 1 | Est. 1 (sd) | True 2 | Est. 2 (sd) |
|------|------|--------|-------------|--------|-------------|
| 1    | 1    | 0.15   | 0.08 (0.03) | 0.70   | 0.73 (0.03) |
|      | 2    | 0.25   | 0.30 (0.03) | 0.20   | 0.16 (0.02) |
|      | 3    | 0.60   | 0.62 (0.04) | 0.10   | 0.11 (0.02) |
| 2    | 1    | 0.20   | 0.23 (0.03) | 0.55   | 0.55 (0.03) |
|      | 2    | 0.35   | 0.32 (0.03) | 0.30   | 0.29 (0.03) |
|      | 3    | 0.45   | 0.45 (0.04) | 0.15   | 0.15 (0.02) |
| 3    | 1    | 0.10   | 0.13 (0.03) | 0.80   | 0.82 (0.03) |
|      | 2    | 0.15   | 0.15 (0.03) | 0.15   | 0.13 (0.02) |
|      | 3    | 0.75   | 0.72 (0.04) | 0.05   | 0.05 (0.02) |
| 4    | 1    | 0.25   | 0.26 (0.03) | 0.45   | 0.44 (0.03) |
|      | 2    | 0.40   | 0.35 (0.04) | 0.35   | 0.37 (0.03) |
|      | 3    | 0.35   | 0.40 (0.04) | 0.20   | 0.19 (0.02) |
| 5    | 1    | 0.40   | 0.40 (0.04) | 0.40   | 0.44 (0.03) |
|      | 2    | 0.50   | 0.52 (0.04) | 0.50   | 0.46 (0.03) |
|      | 3    | 0.10   | 0.08 (0.02) | 0.10   | 0.11 (0.02) |
| 6    | 1    | 0.70   | 0.74 (0.04) | 0.70   | 0.72 (0.03) |
|      | 2    | 0.10   | 0.07 (0.02) | 0.10   | 0.09 (0.02) |
|      | 3    | 0.20   | 0.18 (0.03) | 0.20   | 0.19 (0.02) |
| 7    | 1    | 0.10   | 0.08 (0.02) | 0.10   | 0.09 (0.02) |
|      | 2    | 0.50   | 0.50 (0.04) | 0.50   | 0.52 (0.03) |
|      | 3    | 0.40   | 0.42 (0.04) | 0.40   | 0.39 (0.03) |
| 8    | 1    | 0.33   | 0.29 (0.03) | 0.33   | 0.32 (0.03) |
|      | 2    | 0.33   | 0.34 (0.04) | 0.33   | 0.34 (0.03) |
|      | 3    | 0.33   | 0.34 (0.04) | 0.33   | 0.34 (0.03) |

**Table S2.** Estimated values of the item probability parameters for the item selection model for simulation 1.

| Variable    | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 8    |
|-------------|------|------|------|------|------|------|------|------|
| PIP (item)  | 1.00 | 1.00 | 1.00 | 1.00 | 0.08 | 0.02 | 0.03 | 0.04 |
| PIP (simul) | 1.00 | 1.00 | 1.00 | 1.00 | 0.07 | 0.02 | 0.03 | 0.04 |

**Table S3.** Posterior inclusion probabilities (PIP) for each item variable in the LCR model for simulation 1. PIP (item): item selection alone; PIP (simul): simultaneous selection.



**Figure S1** Ridgeline plots of posterior distribution of regression coefficients for simulation 1.

## Simulation 2: Parameter Estimates and Selection Results

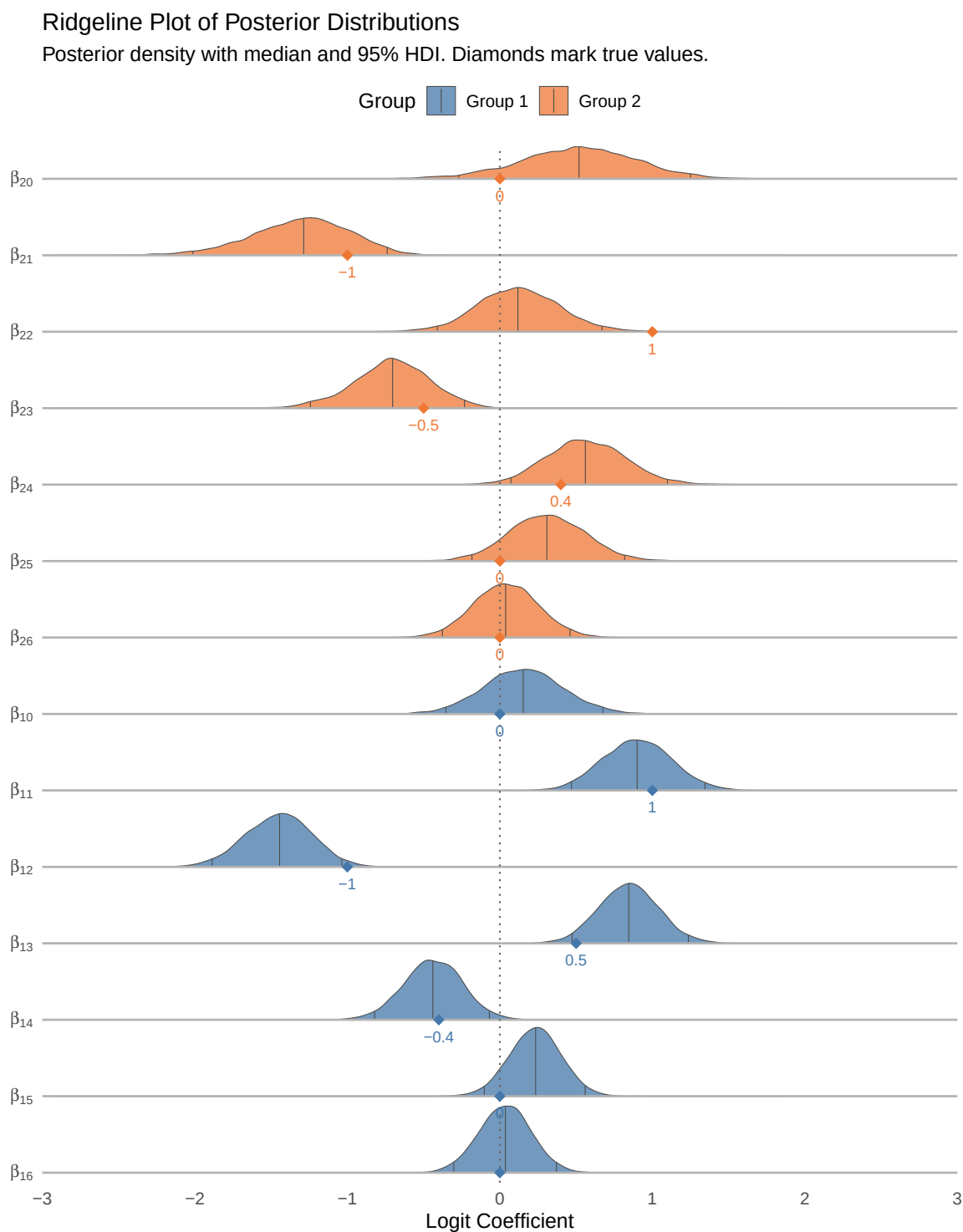
Table S4 contains the true and estimated values of the item probability parameters for the collapsed LCR model for simulation 2, computed post-hoc. Table S5 contains posterior inclusion probabilities for each item variable in simulation 2. Figure S2 contains a ridgeline plot illustrating the posterior distributions of the regression coefficients for the LCR model with item selection only applied to simulation 2.

| Var. | Cat. | True 1 | Est. 1 (sd) | True 2 | Est. 2 (sd) | True 3 | Est. 3 (sd) |
|------|------|--------|-------------|--------|-------------|--------|-------------|
| 1    | 1    | 0.15   | 0.18 (0.03) | 0.60   | 0.58 (0.04) | 0.80   | 0.80 (0.05) |
|      | 2    | 0.85   | 0.82 (0.03) | 0.40   | 0.42 (0.04) | 0.20   | 0.20 (0.05) |
| 2    | 1    | 0.25   | 0.24 (0.03) | 0.45   | 0.45 (0.04) | 0.70   | 0.77 (0.06) |
|      | 2    | 0.75   | 0.76 (0.03) | 0.55   | 0.55 (0.04) | 0.30   | 0.23 (0.06) |
| 3    | 1    | 0.70   | 0.70 (0.04) | 0.20   | 0.23 (0.03) | 0.65   | 0.71 (0.08) |
|      | 2    | 0.30   | 0.30 (0.04) | 0.80   | 0.77 (0.03) | 0.35   | 0.29 (0.08) |
| 4    | 1    | 0.10   | 0.08 (0.02) | 0.35   | 0.39 (0.04) | 0.70   | 0.64 (0.07) |
|      | 2    | 0.25   | 0.28 (0.03) | 0.40   | 0.34 (0.04) | 0.20   | 0.26 (0.06) |
|      | 3    | 0.65   | 0.64 (0.04) | 0.25   | 0.27 (0.03) | 0.10   | 0.09 (0.04) |
| 5    | 1    | 0.20   | 0.18 (0.03) | 0.25   | 0.29 (0.04) | 0.65   | 0.66 (0.07) |
|      | 2    | 0.15   | 0.17 (0.03) | 0.60   | 0.59 (0.04) | 0.25   | 0.25 (0.06) |
|      | 3    | 0.65   | 0.65 (0.04) | 0.15   | 0.12 (0.03) | 0.10   | 0.09 (0.04) |
| 6    | 1    | 0.15   | 0.11 (0.03) | 0.50   | 0.45 (0.03) | 0.75   | 0.79 (0.06) |
|      | 2    | 0.20   | 0.21 (0.03) | 0.35   | 0.38 (0.04) | 0.15   | 0.13 (0.05) |
|      | 3    | 0.65   | 0.68 (0.04) | 0.15   | 0.16 (0.03) | 0.10   | 0.09 (0.03) |
| 7    | 1    | 0.10   | 0.10 (0.02) | 0.25   | 0.25 (0.03) | 0.60   | 0.51 (0.06) |
|      | 2    | 0.15   | 0.13 (0.03) | 0.35   | 0.33 (0.03) | 0.25   | 0.28 (0.06) |
|      | 3    | 0.25   | 0.23 (0.03) | 0.25   | 0.25 (0.03) | 0.10   | 0.15 (0.04) |
|      | 4    | 0.50   | 0.53 (0.04) | 0.15   | 0.17 (0.03) | 0.05   | 0.06 (0.03) |
| 8    | 1    | 0.15   | 0.20 (0.03) | 0.20   | 0.20 (0.03) | 0.55   | 0.67 (0.07) |
|      | 2    | 0.20   | 0.14 (0.03) | 0.45   | 0.45 (0.04) | 0.20   | 0.16 (0.06) |
|      | 3    | 0.20   | 0.20 (0.03) | 0.25   | 0.26 (0.03) | 0.15   | 0.12 (0.04) |
|      | 4    | 0.45   | 0.46 (0.04) | 0.10   | 0.10 (0.02) | 0.10   | 0.05 (0.03) |
| 9    | 1    | 0.40   | 0.35 (0.04) | 0.40   | 0.43 (0.04) | 0.40   | 0.39 (0.07) |
|      | 2    | 0.50   | 0.53 (0.04) | 0.50   | 0.46 (0.04) | 0.50   | 0.47 (0.07) |
|      | 3    | 0.10   | 0.12 (0.02) | 0.10   | 0.11 (0.02) | 0.10   | 0.14 (0.04) |
| 10   | 1    | 0.70   | 0.72 (0.04) | 0.70   | 0.72 (0.04) | 0.70   | 0.68 (0.08) |
|      | 2    | 0.10   | 0.10 (0.02) | 0.10   | 0.11 (0.02) | 0.10   | 0.15 (0.04) |
|      | 3    | 0.20   | 0.18 (0.03) | 0.20   | 0.17 (0.03) | 0.20   | 0.17 (0.05) |
| 11   | 1    | 0.20   | 0.21 (0.03) | 0.20   | 0.20 (0.03) | 0.20   | 0.20 (0.05) |
|      | 2    | 0.20   | 0.18 (0.03) | 0.20   | 0.20 (0.03) | 0.20   | 0.15 (0.04) |
|      | 3    | 0.20   | 0.22 (0.03) | 0.20   | 0.20 (0.03) | 0.20   | 0.23 (0.05) |
|      | 4    | 0.20   | 0.21 (0.03) | 0.20   | 0.20 (0.03) | 0.20   | 0.20 (0.05) |
|      | 5    | 0.20   | 0.19 (0.03) | 0.20   | 0.20 (0.03) | 0.20   | 0.22 (0.05) |
| 12   | 1    | 0.10   | 0.15 (0.03) | 0.10   | 0.11 (0.02) | 0.10   | 0.08 (0.03) |
|      | 2    | 0.15   | 0.17 (0.03) | 0.15   | 0.13 (0.02) | 0.15   | 0.16 (0.04) |
|      | 3    | 0.20   | 0.16 (0.03) | 0.20   | 0.20 (0.03) | 0.20   | 0.25 (0.05) |
|      | 4    | 0.25   | 0.24 (0.03) | 0.25   | 0.21 (0.03) | 0.25   | 0.21 (0.05) |
|      | 5    | 0.30   | 0.28 (0.03) | 0.30   | 0.34 (0.03) | 0.30   | 0.31 (0.06) |
| 13   | 1    | 0.20   | 0.20 (0.03) | 0.20   | 0.20 (0.03) | 0.20   | 0.23 (0.05) |
|      | 2    | 0.30   | 0.27 (0.03) | 0.30   | 0.27 (0.03) | 0.30   | 0.33 (0.06) |
|      | 3    | 0.30   | 0.31 (0.04) | 0.30   | 0.30 (0.03) | 0.30   | 0.29 (0.06) |
|      | 4    | 0.10   | 0.10 (0.02) | 0.10   | 0.14 (0.03) | 0.10   | 0.09 (0.03) |
|      | 5    | 0.10   | 0.11 (0.02) | 0.10   | 0.09 (0.02) | 0.10   | 0.06 (0.03) |

**Table S4.** True and estimated values of the item probability parameters for the item selection model for dataset 2.

| Variable    | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 8    | 9    | 10   | 11   | 12   | 13   |
|-------------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| PIP (item)  | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 0.01 | 0.00 | 0.00 | 0.00 | 0.00 |
| PIP (simul) | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 0.01 | 0.00 | 0.00 | 0.00 | 0.00 |

**Table S5.** Posterior inclusion probabilities (PIP) for each item variable in the LCR model for dataset 2. PIP (item): item selection alone; PIP (simul): simultaneous selection.



**Figure S2** Ridgeline plots of posterior distribution of regression coefficients for simulation 2.

# CSHQ Sleep Data Analysis

## Study Sample Characteristics

Table S6 contains demographic and diagnostic information for the sleep patterns dataset.

| Demographic                |              | <i>n</i> | %            |
|----------------------------|--------------|----------|--------------|
| Gender                     | Female       | 69       | 46.6         |
|                            | Male         | 79       | 53.4         |
| Diagnosis                  | Neurotypical | 104      | 70.3         |
|                            | Neurodiverse | 44       | 29.7         |
| Diagnosis/comorbidity (ND) |              | <i>n</i> | % (ND group) |
| ASD                        |              | 19       | 43.18        |
| ID/ASD                     |              | 6        | 13.64        |
| ADHD/ASD                   |              | 5        | 11.36        |
| Dyslexia                   |              | 3        | 6.82         |
| ADHD                       |              | 2        | 4.55         |
| ASD/ADHD/Dyslexia          |              | 1        | 2.27         |
| ID/ASD/ADHD                |              | 1        | 2.27         |
| ASD/GDD                    |              | 1        | 2.27         |
| ADHD/Dyslexia              |              | 1        | 2.27         |
| ID/Dyslexia                |              | 1        | 2.27         |
| ID                         |              | 1        | 2.27         |
| Dyspraxia                  |              | 1        | 2.27         |
| Tourette's                 |              | 1        | 2.27         |
| Brain Abnormality          |              | 1        | 2.27         |

**Table S6.** Demographic and diagnostic summary information for the CSHQ dataset (ND = neurodiverse, ASD = Autism Spectrum Disorder, ID = Intellectual Disability, GDD = Global Developmental Delay)

## Model Parameter Estimates

Table S7 contains estimates for the item probability parameters computed post-hoc for the sleep data LCR model with item selection. Table S8 contains posterior mean values for LCR regression coefficient parameters, along with sd and 95% HDI values. Table S9 contains the estimates for the regression coefficient parameters for the ASD diagnosis variable with predictor selection.

Table S7: Estimated values of the item probability parameters for the item selection model for the CSHQ sleep data

| Variable           | Category |   | Est. 1 (sd) | Est. 2 (sd) | Est. 3 (sd) | Est. 4 (sd) |
|--------------------|----------|---|-------------|-------------|-------------|-------------|
| Bedtime Resistance | 1        | 1 | 0.78 (0.06) | 0.77 (0.09) | 0.63 (0.10) | 0.68 (0.11) |
|                    |          | 2 | 0.18 (0.05) | 0.18 (0.07) | 0.30 (0.09) | 0.27 (0.09) |
|                    |          | 3 | 0.04 (0.03) | 0.05 (0.04) | 0.07 (0.05) | 0.04 (0.04) |
|                    | 2        | 1 | 0.94 (0.03) | 0.75 (0.07) | 0.90 (0.05) | 0.51 (0.11) |
|                    |          | 2 | 0.02 (0.02) | 0.09 (0.05) | 0.07 (0.04) | 0.33 (0.10) |
|                    |          | 3 | 0.05 (0.03) | 0.16 (0.06) | 0.03 (0.03) | 0.16 (0.07) |
|                    | 3        | 1 | 0.94 (0.03) | 0.75 (0.07) | 0.87 (0.06) | 0.43 (0.11) |
|                    |          |   |             |             |             |             |
|                    |          |   |             |             |             |             |

*Continued on next page*

Table S7 – Continued from previous page

| Variable          | Category | Est. 1 (sd) | Est. 2 (sd) | Est. 3 (sd) | Est. 4 (sd) |
|-------------------|----------|-------------|-------------|-------------|-------------|
| Sleep Onset Delay | 5        | 2           | 0.02 (0.02) | 0.07 (0.04) | 0.09 (0.05) |
|                   |          | 3           | 0.05 (0.03) | 0.18 (0.07) | 0.03 (0.03) |
|                   |          | 1           | 0.89 (0.04) | 0.14 (0.07) | 0.77 (0.08) |
|                   |          | 2           | 0.09 (0.04) | 0.38 (0.09) | 0.20 (0.08) |
|                   |          | 3           | 0.02 (0.02) | 0.47 (0.08) | 0.04 (0.04) |
|                   | 2        | 1           | 0.58 (0.06) | 0.48 (0.09) | 0.15 (0.07) |
|                   |          | 2           | 0.27 (0.06) | 0.41 (0.09) | 0.57 (0.09) |
|                   |          | 3           | 0.15 (0.05) | 0.11 (0.05) | 0.28 (0.08) |
|                   |          | 1           | 0.85 (0.05) | 0.65 (0.09) | 0.14 (0.07) |
|                   |          | 2           | 0.12 (0.05) | 0.29 (0.09) | 0.55 (0.09) |
| Sleep Duration    | 3        | 3           | 0.03 (0.02) | 0.06 (0.04) | 0.32 (0.09) |
|                   |          | 1           | 0.86 (0.05) | 0.65 (0.09) | 0.15 (0.07) |
|                   |          | 2           | 0.10 (0.04) | 0.27 (0.09) | 0.53 (0.09) |
|                   |          | 3           | 0.05 (0.03) | 0.09 (0.05) | 0.32 (0.09) |
|                   |          | 1           | 0.86 (0.05) | 0.74 (0.08) | 0.22 (0.08) |
|                   | 2        | 2           | 0.11 (0.04) | 0.23 (0.08) | 0.54 (0.09) |
|                   |          | 3           | 0.03 (0.02) | 0.03 (0.03) | 0.24 (0.08) |
|                   |          | 1           | 0.80 (0.05) | 0.14 (0.07) | 0.89 (0.06) |
|                   |          | 2           | 0.13 (0.04) | 0.22 (0.07) | 0.06 (0.05) |
|                   |          | 3           | 0.06 (0.03) | 0.63 (0.09) | 0.05 (0.04) |
| Sleep Anxiety     | 3        | 1           | 0.70 (0.06) | 0.35 (0.08) | 0.67 (0.09) |
|                   |          | 2           | 0.20 (0.05) | 0.19 (0.07) | 0.12 (0.06) |
|                   |          | 3           | 0.10 (0.04) | 0.46 (0.09) | 0.21 (0.08) |
|                   |          | 1           | 0.89 (0.04) | 0.11 (0.06) | 0.76 (0.08) |
|                   |          | 2           | 0.08 (0.04) | 0.35 (0.08) | 0.20 (0.08) |
|                   | 4        | 3           | 0.02 (0.02) | 0.54 (0.09) | 0.04 (0.04) |
|                   |          | 1           | 0.70 (0.06) | 0.63 (0.08) | 0.41 (0.09) |
|                   |          | 2           | 0.27 (0.06) | 0.26 (0.07) | 0.40 (0.09) |
|                   |          | 3           | 0.03 (0.02) | 0.11 (0.05) | 0.19 (0.07) |
|                   |          | 1           | 0.90 (0.04) | 0.48 (0.09) | 0.71 (0.09) |
| Night Waking      | 2        | 2           | 0.08 (0.04) | 0.34 (0.08) | 0.20 (0.08) |
|                   |          | 3           | 0.02 (0.02) | 0.17 (0.07) | 0.10 (0.07) |
|                   |          | 1           | 0.76 (0.06) | 0.33 (0.08) | 0.31 (0.08) |
|                   |          | 2           | 0.21 (0.06) | 0.46 (0.09) | 0.33 (0.09) |
|                   |          | 3           | 0.03 (0.02) | 0.21 (0.07) | 0.35 (0.09) |
|                   | 3        | 1           | 0.94 (0.03) | 0.64 (0.08) | 0.41 (0.10) |
|                   |          | 2           | 0.04 (0.03) | 0.33 (0.08) | 0.25 (0.08) |
|                   |          | 3           | 0.02 (0.02) | 0.03 (0.03) | 0.33 (0.09) |
|                   |          | 1           | 0.91 (0.06) | 0.87 (0.09) | 0.81 (0.10) |
|                   |          | 2           | 0.05 (0.03) | 0.08 (0.04) | 0.09 (0.05) |
| Parasomnias       | 3        | 3           | 0.05 (0.03) | 0.05 (0.04) | 0.09 (0.05) |
|                   |          | 1           | 0.68 (0.06) | 0.51 (0.09) | 0.77 (0.09) |
|                   |          | 2           | 0.26 (0.06) | 0.46 (0.09) | 0.13 (0.07) |
|                   |          | 3           | 0.05 (0.03) | 0.03 (0.03) | 0.10 (0.06) |
|                   |          | 1           | 0.48 (0.06) | 0.28 (0.08) | 0.27 (0.08) |
|                   | 4        | 2           | 0.43 (0.06) | 0.54 (0.08) | 0.35 (0.09) |
|                   |          | 3           | 0.10 (0.04) | 0.19 (0.07) | 0.38 (0.09) |
|                   |          | 1           | 0.95 (0.05) | 0.91 (0.09) | 0.88 (0.10) |
|                   |          | 2           | 0.05 (0.03) | 0.09 (0.05) | 0.12 (0.06) |
|                   |          |             |             |             | 0.17 (0.08) |

Continued on next page

Table S7 – Continued from previous page

| Variable                   |                    | Category    | Est. 1 (sd) | Est. 2 (sd) | Est. 3 (sd) | Est. 4 (sd) |
|----------------------------|--------------------|-------------|-------------|-------------|-------------|-------------|
| Sleep Disordered Breathing | 5                  | 1           | 0.89 (0.04) | 0.67 (0.09) | 0.75 (0.08) | 0.29 (0.09) |
|                            |                    | 2           | 0.06 (0.03) | 0.18 (0.07) | 0.16 (0.07) | 0.41 (0.10) |
|                            |                    | 3           | 0.06 (0.03) | 0.16 (0.06) | 0.09 (0.06) | 0.30 (0.09) |
|                            | 6                  | 1           | 0.93 (0.05) | 0.90 (0.08) | 0.79 (0.10) | 0.73 (0.11) |
|                            |                    | 2           | 0.07 (0.03) | 0.10 (0.05) | 0.21 (0.08) | 0.27 (0.09) |
|                            | 7                  | 1           | 0.80 (0.06) | 0.69 (0.09) | 0.68 (0.10) | 0.60 (0.11) |
|                            |                    | 2           | 0.19 (0.05) | 0.26 (0.08) | 0.26 (0.08) | 0.35 (0.10) |
|                            |                    | 3           | 0.02 (0.02) | 0.05 (0.04) | 0.06 (0.04) | 0.04 (0.04) |
|                            | 1                  | 1           | 0.82 (0.06) | 0.72 (0.10) | 0.72 (0.10) | 0.62 (0.11) |
|                            |                    | 2           | 0.14 (0.04) | 0.22 (0.07) | 0.16 (0.07) | 0.34 (0.10) |
|                            |                    | 3           | 0.05 (0.03) | 0.05 (0.04) | 0.12 (0.06) | 0.04 (0.04) |
|                            | Daytime Sleepiness | 2           | 1           | 0.98 (0.05) | 0.97 (0.08) | 0.90 (0.09) |
| 2                          |                    |             | 0.02 (0.02) | 0.03 (0.03) | 0.10 (0.05) | 0.17 (0.08) |
| 3                          |                    | 1           | 0.92 (0.05) | 0.94 (0.08) | 0.79 (0.10) | 0.80 (0.11) |
|                            |                    | 2           | 0.08 (0.04) | 0.06 (0.04) | 0.21 (0.08) | 0.20 (0.08) |
| 1                          |                    | 1           | 0.45 (0.07) | 0.66 (0.09) | 0.46 (0.10) | 0.53 (0.11) |
|                            |                    | 2           | 0.35 (0.06) | 0.16 (0.07) | 0.43 (0.10) | 0.30 (0.10) |
|                            |                    | 3           | 0.20 (0.05) | 0.18 (0.07) | 0.11 (0.06) | 0.17 (0.08) |
| 2                          |                    | 1           | 0.68 (0.06) | 0.77 (0.08) | 0.16 (0.07) | 0.22 (0.09) |
|                            |                    | 2           | 0.28 (0.06) | 0.20 (0.08) | 0.63 (0.09) | 0.53 (0.10) |
|                            |                    | 3           | 0.03 (0.02) | 0.03 (0.03) | 0.21 (0.07) | 0.25 (0.09) |
| 3                          |                    | 1           | 0.59 (0.07) | 0.75 (0.09) | 0.30 (0.09) | 0.34 (0.10) |
|                            |                    | 2           | 0.33 (0.06) | 0.22 (0.08) | 0.34 (0.09) | 0.44 (0.10) |
|                            |                    | 3           | 0.08 (0.04) | 0.03 (0.03) | 0.36 (0.07) | 0.21 (0.08) |
| 4                          |                    | 1           | 0.59 (0.06) | 0.75 (0.08) | 0.30 (0.08) | 0.34 (0.10) |
|                            |                    | 2           | 0.37 (0.06) | 0.22 (0.08) | 0.34 (0.09) | 0.44 (0.10) |
|                            |                    | 3           | 0.03 (0.02) | 0.03 (0.03) | 0.36 (0.09) | 0.21 (0.08) |
| 5                          | 1                  | 0.84 (0.05) | 0.81 (0.07) | 0.42 (0.09) | 0.33 (0.10) |             |
|                            | 2                  | 0.14 (0.05) | 0.16 (0.06) | 0.46 (0.09) | 0.55 (0.11) |             |
|                            | 3                  | 0.02 (0.02) | 0.03 (0.03) | 0.12 (0.06) | 0.12 (0.07) |             |
| 6                          | 1                  | 0.62 (0.06) | 0.60 (0.09) | 0.16 (0.07) | 0.09 (0.06) |             |
|                            | 2                  | 0.36 (0.06) | 0.38 (0.09) | 0.51 (0.10) | 0.59 (0.10) |             |
|                            | 3                  | 0.02 (0.02) | 0.03 (0.03) | 0.33 (0.09) | 0.31 (0.10) |             |
| 7                          | 1                  | 0.92 (0.06) | 0.90 (0.09) | 0.78 (0.10) | 0.80 (0.11) |             |
|                            | 2                  | 0.05 (0.03) | 0.05 (0.04) | 0.12 (0.06) | 0.04 (0.04) |             |
|                            | 3                  | 0.03 (0.02) | 0.05 (0.04) | 0.09 (0.05) | 0.16 (0.07) |             |
| 8                          | 1                  | 0.82 (0.06) | 0.77 (0.09) | 0.76 (0.10) | 0.53 (0.11) |             |
|                            | 2                  | 0.17 (0.05) | 0.17 (0.07) | 0.21 (0.08) | 0.36 (0.10) |             |
|                            | 3                  | 0.02 (0.02) | 0.05 (0.04) | 0.03 (0.03) | 0.12 (0.06) |             |

| Group | Parameter                      | Post. Mean | Post. SD | 95% HDI         |
|-------|--------------------------------|------------|----------|-----------------|
| B     | $\beta_{B0}$ (intercept)       | -0.32      | 0.37     | [-1.08, 0.37]   |
|       | $\beta_{B1}$ (age)             | -0.65      | 0.37     | [-1.38, -0.05]  |
|       | $\beta_{B2}$ (gender)          | -0.93      | 0.67     | [-2.33, 0.34]   |
|       | $\beta_{B3}$ (ASD diagnosis)   | 1.88       | 1.08     | [-0.09, 4.10]   |
|       | $\beta_{B4}$ (ID diagnosis)    | -2.72      | 3.65     | [-10.38, 3.62]  |
|       | $\beta_{B5}$ (other diagnoses) | -5.39      | 3.23     | [-11.97, -0.19] |
| C     | $\beta_{C0}$ (intercept)       | -1.45      | 0.44     | [-2.29, -0.58]  |
|       | $\beta_{C1}$ (age)             | 0.24       | 0.33     | [-0.39, 0.91]   |
|       | $\beta_{C2}$ (gender)          | 0.02       | 0.70     | [-1.32, 1.38]   |
|       | $\beta_{C3}$ (ASD diagnosis)   | 2.88       | 0.93     | [1.07, 4.71]    |
|       | $\beta_{C4}$ (ID diagnosis)    | 2.27       | 1.75     | [-0.92, 5.75]   |
|       | $\beta_{C5}$ (other diagnoses) | -0.55      | 1.07     | [-2.52, 1.61]   |
| D     | $\beta_{D0}$ (intercept)       | -1.60      | 0.33     | [-2.26, -0.94]  |
|       | $\beta_{D1}$ (age)             | -0.12      | 0.27     | [-0.65, 0.41]   |
|       | $\beta_{D2}$ (gender)          | 0.02       | 0.60     | [-1.16, 1.19]   |
|       | $\beta_{D3}$ (ASD diagnosis)   | 3.40       | 0.85     | [1.81, 5.09]    |
|       | $\beta_{D4}$ (ID diagnosis)    | 0.53       | 1.79     | [-2.75, 4.20]   |
|       | $\beta_{D5}$ (other diagnoses) | -2.05      | 0.94     | [-2.90, 0.77]   |

**Table S8.** Summary table containing the posterior mean estimated from MCMC samples for the CSHQ data using only item selection, along with the 95% HDI and posterior SD. The gender variable was binary encoded, taking value 1 for a response of female.

| Group | Parameter                    | Post. Mean | Post. SD | 95% HDI        |
|-------|------------------------------|------------|----------|----------------|
| B     | $\beta_{B0}$ (intercept)     | -0.60      | 0.24     | [-1.06, -0.16] |
|       | $\beta_{B3}$ (ASD diagnosis) | 0.82       | 0.89     | [-0.98, 2.51]  |
| C     | $\beta_{C0}$ (intercept)     | -1.33      | 0.35     | [-2.01, -0.63] |
|       | $\beta_{C3}$ (ASD diagnosis) | 2.96       | 0.89     | [1.22, 4.69]   |
| D     | $\beta_{D0}$ (intercept)     | -1.67      | 0.37     | [-2.42, -0.97] |
|       | $\beta_{D3}$ (ASD diagnosis) | 2.30       | 0.87     | [1.39, 4.82]   |

**Table S9.** Summary table containing the posterior mean estimated from MCMC samples for the CSHQ data using full variable selection, along with the 95% HDI and posterior SD. Estimates are computed over samples taken from iterations when the ASD diagnosis variable was included in the model.

## Variable Selection Results

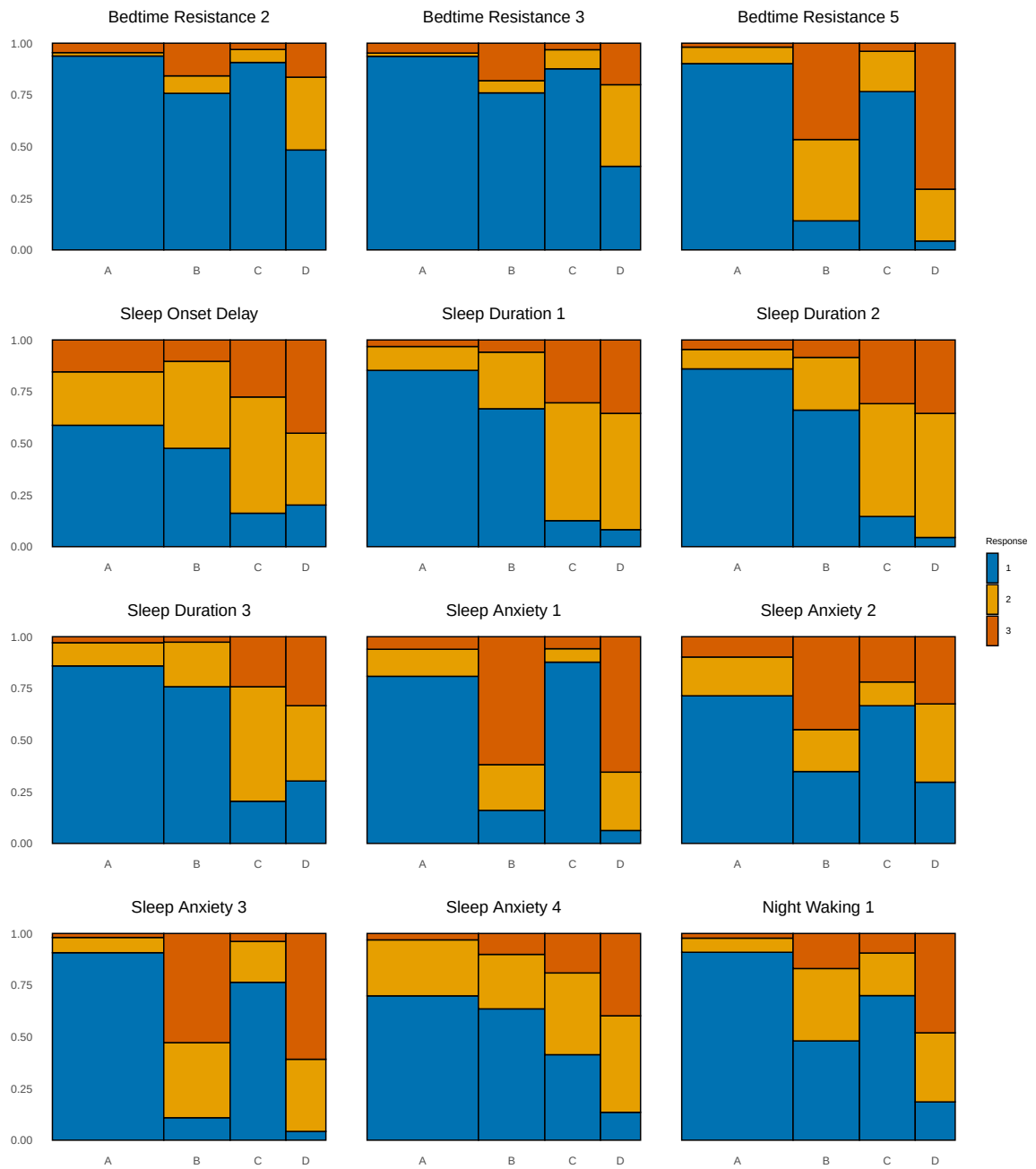
Table S10 contains posterior inclusion probabilities for each item variable in the CSHQ survey.

| CSHQ Subscale              | Variable | PIP (LCR - Item) | PIP (LCR - simul.) | PIP (LCA) |
|----------------------------|----------|------------------|--------------------|-----------|
| Bedtime Resistance         | 1        | 0.00             | 0.00               | 0.00      |
|                            | 2        | 0.96             | 0.99               | 1.00      |
|                            | 3        | 1.00             | 1.00               | 1.00      |
|                            | 5        | 1.00             | 1.00               | 1.00      |
| Sleep Onset Delay          |          | 1.00             | 1.00               | 1.00      |
| Sleep Duration             | 1        | 1.00             | 1.00               | 1.00      |
|                            | 2        | 1.00             | 1.00               | 1.00      |
|                            | 3        | 1.00             | 1.00               | 1.00      |
| Sleep Anxiety              | 1        | 1.00             | 1.00               | 1.00      |
|                            | 2        | 0.99             | 1.00               | 1.00      |
|                            | 3        | 1.00             | 1.00               | 1.00      |
|                            | 4        | 1.00             | 1.00               | 1.00      |
| Night Waking               | 1        | 1.00             | 1.00               | 1.00      |
|                            | 2        | 1.00             | 1.00               | 1.00      |
|                            | 3        | 1.00             | 1.00               | 1.00      |
| Parasomnias                | 1        | 0.00             | 0.00               | 0.00      |
|                            | 2        | 0.50             | 0.72               | 0.91      |
|                            | 3        | 0.99             | 1.00               | 1.00      |
|                            | 4        | 0.02             | 0.02               | 0.00      |
|                            | 5        | 1.00             | 1.00               | 1.00      |
|                            | 6        | 0.28             | 0.27               | 0.45      |
|                            | 7        | 0.00             | 0.00               | 0.00      |
| Sleep Disordered Breathing | 1        | 0.01             | 0.00               | 0.00      |
|                            | 2        | 0.08             | 0.09               | 0.01      |
|                            | 3        | 0.13             | 0.08               | 0.01      |
| Daytime Sleepiness         | 1        | 0.15             | 0.19               | 0.12      |
|                            | 2        | 1.00             | 1.00               | 1.00      |
|                            | 3        | 0.48             | 0.42               | 0.29      |
|                            | 4        | 1.00             | 1.00               | 1.00      |
|                            | 5        | 1.00             | 1.00               | 1.00      |
|                            | 6        | 1.00             | 1.00               | 1.00      |
|                            | 7        | 0.00             | 0.00               | 0.00      |
|                            | 8        | 0.01             | 0.00               | 0.00      |

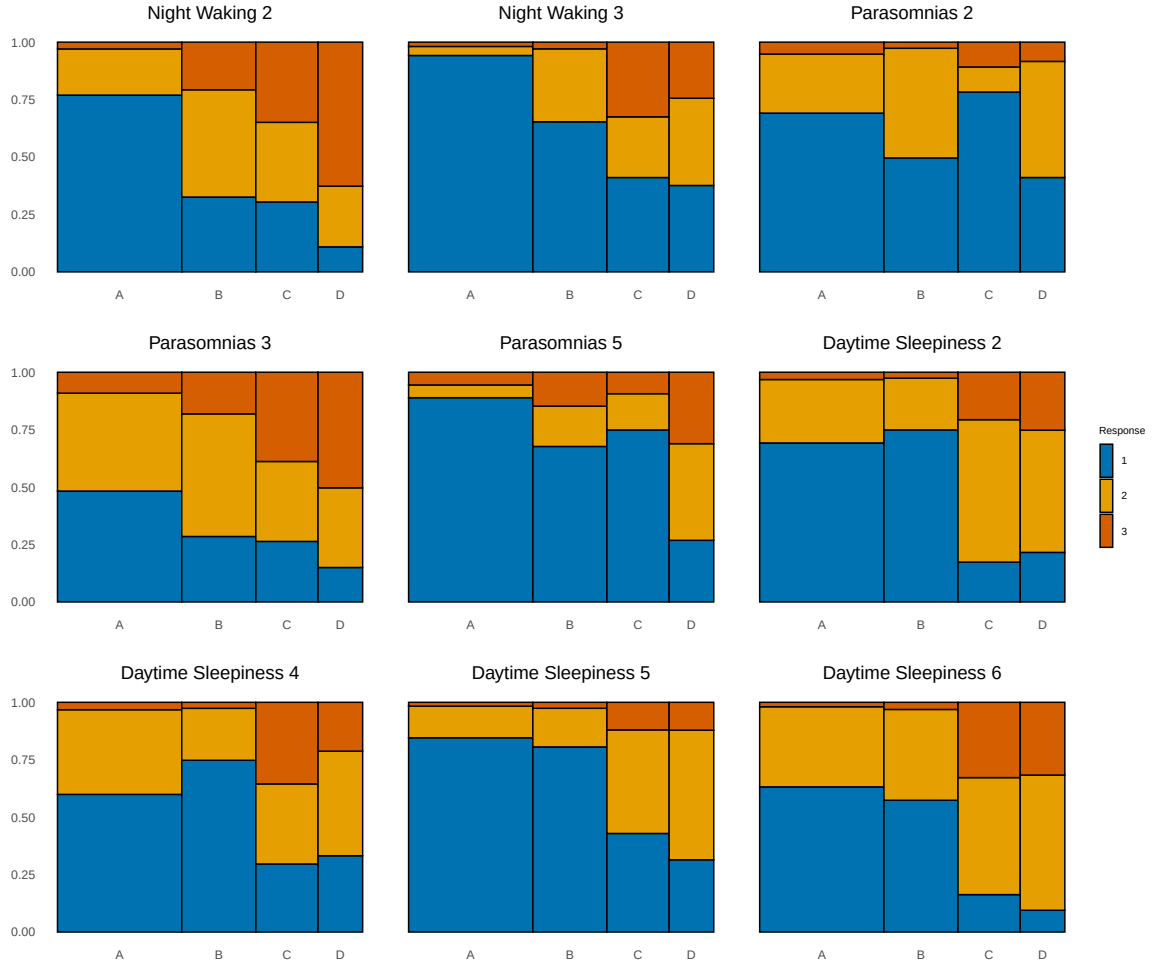
**Table S10.** Table containing the posterior inclusion probabilities (PIP) for each of the CSHQ items, from the item selection model for latent class regression (LCR - Item), the simultaneous item variable and predictor variable selection (LCR - simul.) and from the collapsed latent class analysis model (LCA)

## Supplementary Visualisations

Figures S3 and S4 contain mosaic plots for the each of the item variables of the CSHQ sleep patterns dataset which were retained by the collapsed sampler based on the posterior inclusion probabilities being greater than 0.5.



**Figure S3** Mosaic plots from item probabilities computed post-hoc, and baseline class membership probabilities for the variable selection LCR model, including only variables with PIP > 0.5



**Figure S4** Mosaic plots from item probabilities computed post-hoc, and baseline class membership probabilities for the variable selection LCR model, including only variables with PIP > 0.5

## Technical Derivations

### Derivation of Pólya-Gamma Sampling Method for LCR

This follows the method of Polson et al. (2013). Consider the full conditional for  $\beta$  from (4). We can consider each of the  $\beta_g$  parameters conditional on  $\beta_{-g}$ , the  $\beta$  matrix with the  $g$ -th column removed. We write the expression for the full conditional distribution of  $\beta_g$  as

$$p(\beta_g | \beta_{-g}, \mathbf{X}, \mathbf{Y}) \propto p(\beta_g) \prod_{i=1}^n \left[ \frac{\exp(\mathbf{X}_i^T \beta_g)}{\sum_{h=1}^G \exp(\mathbf{X}_i^T \beta_h)} \right]^{z_{ig}} \left[ 1 - \frac{\exp(\mathbf{X}_i^T \beta_g)}{\sum_{h=1}^G \exp(\mathbf{X}_i^T \beta_h)} \right]^{1-z_{ig}}. \quad (\text{S1})$$

This can be rewritten in the form

$$p(\beta_g | \beta_{-g}, \mathbf{X}, \mathbf{Z}) \propto p(\beta_g) \prod_{i=1}^n \frac{\exp(\eta_{ig})^{z_{ig}}}{1 + \exp(\eta_{ig})},$$

where

$$\eta_{ig} = \mathbf{X}_i^T \boldsymbol{\beta}_g - C_{ig}, \quad C_{ig} = \log \left( \sum_{h \neq g} \exp(\mathbf{X}_i^T \boldsymbol{\beta}_h) \right).$$

By utilising the result from (5), we can introduce the Pólya-Gamma random variables  $\omega_{ig} \sim \text{PG}(1, 0)$  for each  $i = 1, \dots, n, g = 1, \dots, G - 1$ , which allows us to rewrite the full conditional as

$$p(\boldsymbol{\beta}_g \mid \boldsymbol{\beta}_{-g}, \mathbf{X}, \mathbf{Y}) \propto p(\boldsymbol{\beta}_g) \prod_{i=1}^n \exp \left( \kappa_{ig} \eta_{ig} - \frac{\omega_{ig} \eta_{ig}^2}{2} \right),$$

where  $\kappa_{ig} = z_{ig} - \frac{1}{2}$ . We can recognise the second multiplicative term here as a product of Gaussian kernels in the  $\eta$  variables. By setting the prior  $\boldsymbol{\beta}_g \sim \mathcal{N}(\mathbf{m}_{0g}, \mathbf{V}_{0g})$ , we are left with a two part update for  $\boldsymbol{\beta}_g$  and  $\omega_{ig}$ , given by

$$\begin{aligned} \boldsymbol{\beta}_g \mid \boldsymbol{\beta}_{-g}, \omega_{\cdot g}, \mathbf{X}, \mathbf{Z} &\sim \mathcal{N}(\mathbf{m}_g, \mathbf{V}_g), \\ \omega_{ig} \mid \boldsymbol{\beta}, \mathbf{X}_i &\sim \text{PG}(1, \eta_{ig}). \end{aligned} \tag{S2}$$

where

$$\mathbf{V}_g^{-1} = \mathbf{X}^T \Omega_g \mathbf{X} + \mathbf{V}_{0g}^{-1}, \quad \mathbf{m}_g = \mathbf{V}_g \left( \mathbf{X}^T (\boldsymbol{\kappa}_g + \Omega_g \mathbf{C}_{\cdot g}) + \mathbf{V}_{0g}^{-1} \mathbf{m}_{0g} \right), \quad \Omega_g = \text{diag}(\{\omega_{ig}\}_{i=1}^n),$$

and  $\mathbf{C}_{\cdot g}$  is the  $g$ -th column of  $\mathbf{C} = (C_{ig})$ .

## References

N. G. Polson, J. G. Scott, and J. Windle. Bayesian Inference for Logistic Models Using Pólya–Gamma Latent Variables. *Journal of the American Statistical Association*, 108(504):1339–1349, 2013. ISSN 0162-1459. doi: 10.1080/01621459.2013.829001. URL <https://www.jstor.org/stable/24247065>.